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International AS · Edexcel · Chemistry

38 mins   3 questions

Structured Questions

Covalent Bonding & Structure

Evidence of Covalent Bonding / Covalent Bonding / Carbon Allotropes / Electronegativity / VSEPR Theory / Predicting Shapes & Angles

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Total Marks **/38**

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1 (a) This question is about the chlorides of beryllium and calcium.

Complete the electronic configurations of the atoms of beryllium and calcium using the s, p, d notation.

Be $1s^2$

Ca $1s^2$

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(2 marks)

(b) In the gaseous state, beryllium chloride is molecular.

Draw a dot-and-cross diagram to show the bonding in a molecule of beryllium chloride, $BeCl_2$.

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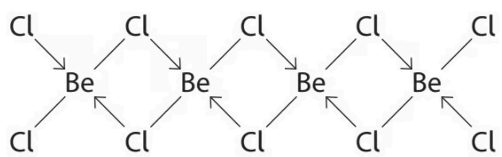
(2 marks)

(c) In the solid state, beryllium chloride forms a polymeric structure.

The diagram shows part of this structure.



The diagram uses lines and arrows to represent the two different types of covalent bond.

Describe how each type of bond is formed.

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(2 marks)

(d) The Cl—Be—Cl bond angle is different in the two forms of beryllium chloride.

Predict the two bond angles, justifying your answers by referring to electron-pair repulsion theory.

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(4 marks)

(e) Anhydrous calcium chloride is a crystalline, ionic solid which melts at 772°C.

Draw a dot-and-cross diagram for calcium chloride.

Show the outer electrons only.

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(2 marks)

(f) Explain why gaseous beryllium chloride and solid calcium chloride have different types of bonding.

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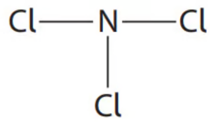
(3 marks)

2 (a) This question is about the structure and bonding of Group 5 chlorides.

Nitrogen trichloride, NCl_3, has a molecular structure.

The displayed formula of a molecule of NCl_3 is shown.



Complete the table for this molecule.

Number of bond pairs around N atom	
Number of lone pairs around N atom	
Cl-N-Cl bond angle	
Name of shape of molecule	

(3 marks)

(b) Under standard conditions, phosphorus(V) chloride (PCl_5) is a solid made up of PCl_4^+ cations and PCl_6^- anions.

Antimony(V) chloride ($SbCl_5$) is a liquid made up of $SbCl_5$ molecules.

i) Explain why PCl_5 has a higher melting temperature than $SbCl_5$.

(2)

ii) Draw a dot-and-cross diagram to show the bonding in a molecule of $SbCl_5$.
Use dots (•) to represent the Sb electrons, and crosses (**x**) to represent the Cl electrons. Show outer electrons only.

(2)

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(4 marks)

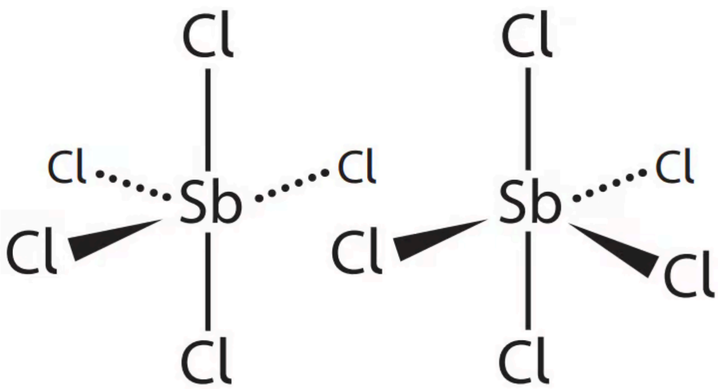
(c) At low temperatures, $SbCl_5$ converts to Sb_2Cl_{10} which contains dative covalent bonds.

i) State what is meant by the term dative covalent bond.

(1)

ii) Complete the diagram to show the dative covalent bonds in Sb_2Cl_{10} .

(1)



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(2 marks)

(d) Arsenic also forms a pentachloride with the formula $AsCl_5$.

Give **one** possible reason why nitrogen is the only Group 5 element that does **not** form a pentachloride.

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(1 mark)

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3 (a) This question is about the bonding in the elements of Period 3 in the Periodic Table. The melting temperatures of the Period 3 elements are shown in the table.

Element	**Na**	**Mg**	**Al**	**Si**	**P**	**S**	**Cl**	**Ar**
Melting temperature / °C	98	650	660	1423	44	120	−101	−189

Sodium, magnesium and aluminium are metals.

i) State what is meant by metallic bonding.

(1)

ii) Melting temperature depends on the strength of metallic bonding.

Explain why the metallic bonding in magnesium is much stronger than that in sodium.

(3)

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(4 marks)

(b) i) In the elements silicon, phosphorus, sulfur and chlorine, the atoms are joined by covalent bonds.

Describe the attraction between the atoms in a covalent bond.

(1)

ii) Explain why the melting temperature of silicon is much higher than that of phosphorus, by referring to their structures.

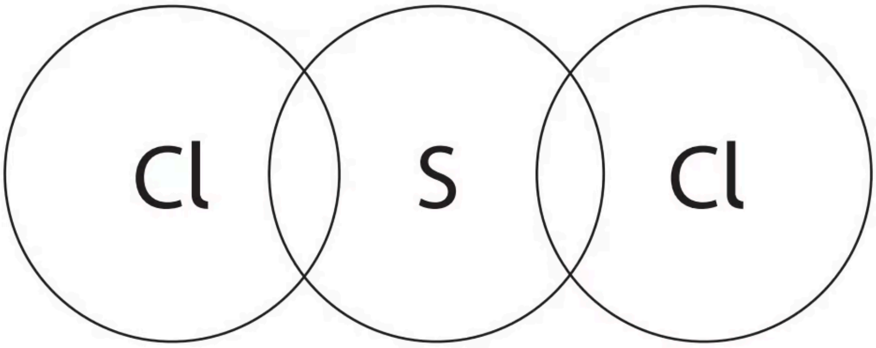
(3)

(4 marks)

(c) Sulfur reacts with chlorine to form sulfur dichloride, SCl_2.

i) Complete the dot-and-cross diagram of a molecule of sulfur dichloride.
Use dots (•) for the chlorine electrons and crosses (×) for the sulfur electrons.
Show the outer shell electrons only.

(2)



ii) Suggest a value for the Cl–S–Cl bond angle. Justify your answer.

(3)

(5 marks)

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